

Clinical Trials and Observations: Monoclonal gammopathy of undetermined significance (MGUS) consistently precedes multiple myeloma: a prospective study

Monoclonal gammopathy of undetermined significance (MGUS) is a premalignant plasma-cell proliferative disorder associated with a life-long risk of progression to multiple myeloma (MM). It is not known whether MM is always preceded by a premalignant asymptomatic MGUS stage. Among 77 469 healthy adults enrolled in the nationwide population-based prospective Prostate, Lung, Colorectal, and Ovarian (PLCO) Cancer Screening Trial, we identified 71 subjects who developed MM during the course of the study in whom serially collected (up to 6) prediagnostic serum samples obtained 2 to 9.8 years prior to MM diagnosis were available. Using assays for monoclonal (M)-proteins (electrophoresis/immunofixation) and kappa-lambda free light chains (FLCs), we determined longitudinally the prevalence of MGUS and characterized patterns of monoclonal immunoglobulin abnormalities prior to MM diagnosis. MGUS was present in 100.0% (87.2%-100.0%), 98.3% (90.8%-100.0%), 97.9% (88.9%-100.0%), 94.6% (81.8%-99.3%), 100.0% (86.3%-100.0%), 93.3% (68.1%-99.8%), and 82.4% (56.6%-96.2%) at 2, 3, 4, 5, 6, 7, and 8+ years prior to MM diagnosis, respectively. In approximately half the study population, the M-protein concentration and involved FLC-ratio levels showed a yearly increase prior to MM diagnosis. In the present study, an asymptomatic MGUS stage consistently preceded MM. Novel molecular markers are needed to better predict progression to MM in patients with MGUS.